Parallel Transport and Curvature

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1 Motivation

Throughout this lecture a connection ∇ is assumed to have been chosen on the smooth manifold $(M, \mathcal{O}, \mathcal{A})$. The connection is the extra structure that defines what it means for a vector to be "constant" as one moves along a curve.

Intuition (the sphere). Take a vector on a sphere and carry it along a path by always keeping it "pointing in the same direction" relative to the path (for example along a north-going segment, then along a segment at constant latitude, then back). Returning to the starting point along different routes, one finds different resulting vectors. This is the effect of curvature: how a vector changes under transport encodes the geometry of the space. This is why the covariant derivative must contain the information we intuitively call curvature.

Connection versus topology. A sphere and an irregularly shaped potato are the same topological manifold (indeed diffeomorphic as smooth manifolds), but with different connections they have different curvature. It is the choice of connection that gives the smooth manifold a geometric "shape". This also lets us ask a physical question: how does a free particle move in a curved space? It moves along the straightest possible lines for the given connection, which we define without any notion of distance or metric.

2 Parallel transport

Definition 1 (Parallel transport)**.** Let $\gamma : I \to M$ be a smooth curve with velocity (tangent) vector $\dot{\gamma}$, and let X be a vector field (or a vector field defined along γ). X is *parallel transported along* γ if

$$\nabla_{\dot{\gamma}} X = 0$$

at every point $\gamma(\lambda)$ of the curve.

So the instruction is: move the vector along the curve such that its covariant derivative in the direction of the curve vanishes. This is a weaker requirement than X being parallel everywhere: it only involves the values of X along the curve. Because $\nabla_X T$ depends only on the value of X at a point (Lecture 7), the condition only needs $\dot{\gamma}$ at each point.

In a chart. With $\nabla_X Y$ in components from Lecture 7, $(\nabla_X Y)^i = X(Y^i) + \Gamma^i{}_{jm} X^m Y^j$, and $\dot{\gamma}^m = (x^m \circ \gamma)'$, parallel transport of X along γ reads

$$\dot{X}^i + \Gamma^i{}_{jk}\, \dot{\gamma}^k\, X^j = 0,$$

where $\dot{X}^i$ is the derivative of the component $X^i(\gamma(\lambda))$.

Parallel transport defines "constant". One is used to thinking of parallel transport as keeping the components constant, but that logic is upside down: the connection *defines* what "constant" means. In a flat Euclidean plane with the standard connection (all $\Gamma^i{}_{jk} = 0$ in Cartesian coordinates) parallel transport keeps the Cartesian components constant, matching intuition. With a different connection the same prescription gives a transported vector that does not look parallel to a Euclidean eye. The parallel transport of a vector from one point to another depends in general on the *path*, and the purpose of the covariant derivative is to calculate systematically the change of a field along a curve.

3 Auto-parallel curves

Definition 2 (Auto-parallel curve)**.** A curve γ is *auto-parallel* (a *geodesic* in the metric context) if its own tangent vector is parallel transported along it:

$$\nabla_{\dot\gamma}\dot\gamma = 0.$$

This is the generalization of a straight line: the velocity does not "turn" with respect to the connection. No metric and no measure of distance is needed. It is also the geometric formulation of Newton's first law: a force-free particle moves along an auto-parallel curve (uniform straight motion), with the connection supplying the geometric framework.

The equation in a chart. Expand $\dot\gamma = \dot\gamma^i\,\partial/\partial x^i$ in a chart (U, x) and use the chain rule:

$$\boxed{\ddot\gamma^i + \Gamma^i{}_{jk}\,\dot\gamma^j\,\dot\gamma^k = 0}$$

where $\gamma^i = x^i \circ \gamma$ and dots denote derivatives with respect to the curve parameter. This is a second-order ODE system for $\gamma^i(\lambda)$.

Affine parameters. Suppose γ is auto-parallel and we re-parameterize, $\tilde\gamma(\mu) = \gamma(\lambda(\mu))$. Then $\tilde\gamma' = \lambda'\,\dot\gamma$ and

$$\nabla_{\tilde\gamma'}\tilde\gamma' = (\lambda')^2\,\nabla_{\dot\gamma}\dot\gamma + \lambda''\lambda'\,\dot\gamma.$$

So $\tilde\gamma$ is again auto-parallel if and only if $\lambda'' = 0$ (for a curve with non-vanishing velocity), i.e. $\lambda = a\mu + b$ with constants $a \neq 0$, b. Hence the equation singles out the parameter up to *affine* transformations (constant rescaling and constant offset); a non-linear re-parameterization introduces an extra term proportional to $\dot\gamma$. In physics, the affine parameter of the curve of a massive particle corresponds to proper time (up to such a rescaling).

Physical interpretation. In Newtonian terms, gravity looks like a force that deflects particles. In the geometric language, a particle "under gravity" simply follows an auto-parallel curve of a curved connection: curvature can mimic a force. This applies to massive particles and to light alike: free motion is along the straightest possible path allowed by the geometry.

3.1 Examples

Example 1 (Flat plane)**.** On $\mathbb{R}^2$ with the Euclidean connection and Cartesian coordinates, all $\Gamma^i{}_{jk} = 0$, so the equation becomes $\ddot\gamma^i = 0$: straight lines traversed with linear parameter. In polar coordinates the coefficients are non-zero and extra terms appear in the equation to account for the turning of the coordinate axes, but the solutions are the same straight lines.

Example 2 (Round sphere)**.** Use spherical coordinates (θ, φ), $\theta \in (0, \pi)$, $\varphi \in (0, 2\pi)$, which cover the sphere except for the poles and a meridian. For the standard round connection the non-zero coefficients are

$$\Gamma^{\theta}{}_{\varphi\varphi} = -\sin\theta\cos\theta, \qquad \Gamma^{\varphi}{}_{\theta\varphi} = \Gamma^{\varphi}{}_{\varphi\theta} = \cot\theta.$$

The auto-parallel equations are

$$\ddot{\theta} - \sin\theta\cos\theta\,\dot{\varphi}^{\,2} = 0, \qquad \ddot{\varphi} + 2\cot\theta\,\dot{\theta}\,\dot{\varphi} = 0.$$

For $\theta \equiv \pi/2$ the first equation holds identically and the second gives $\ddot{\varphi} = 0$: motion along the equator with constant angular velocity is a solution. In general the solutions are *great circles*. To cover the whole sphere one uses overlapping charts, with the coefficients related by the transformation law of Lecture 7.

4 Torsion

The connection coefficients are not the components of a tensor, but suitable combinations of them are.

Definition 3 (Torsion)**.** For vector fields X, Y define

$$T(X, Y) := \nabla_X Y - \nabla_Y X - [X, Y],$$

where the *commutator* $[X, Y]$ is the vector field defined by $[X, Y](f) = X(Y(f)) - Y(X(f))$.

The commutator measures how much the two vector fields fail to commute as derivations on functions. T is $C^\infty(M)$-linear in both slots, so it is a tensor field (a $(1, 2)$ tensor, once a covector ω is included in $T(\omega, X, Y) = \omega\big(T(X, Y)\big)$).

Check of C^∞-linearity. Replace Y by fY with $f \in C^\infty(M)$. By the Leibniz rule, $\nabla_X(fY) = X(f)\,Y + f\nabla_X Y$, $\nabla_{fY} X = f\nabla_Y X$ and $[X, fY] = X(f)\,Y + f[X, Y]$. Hence the terms containing derivatives of f cancel,

$$T(X, fY) = X(f)Y + f\nabla_X Y - f\nabla_Y X - X(f)Y - f[X, Y] = f\,T(X, Y),$$

and similarly in the first slot (using $T(X, Y) = -T(Y, X)$).

Components. In a chart, $[\partial_j, \partial_k] = 0$, so $T(\partial_j, \partial_k) = \nabla_{\partial_j}\partial_k - \nabla_{\partial_k}\partial_j$ and with the convention of Lecture 7,

$$T^i{}_{jk} = \Gamma^i{}_{kj} - \Gamma^i{}_{jk},$$

i.e. the antisymmetric part of the connection coefficients. Being a tensor, if it vanishes in one chart it vanishes in all. A clever choice of chart can remove the symmetric part of Γ at a point (normal coordinates, Lecture 7) but never the antisymmetric part.

In this course we restrict to *torsion-free* connections, $T = 0$. These are the ones needed for general relativity, where the connection is determined from a metric.

5 Curvature

Definition 4 (Riemann curvature)**.** The *Riemann curvature* of a connection ∇ is the map that takes vector fields X, Y, Z to the vector field

$$R(Z; X, Y) := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z,$$

and as a $(1, 3)$ tensor field $\mathrm{Riem}(\omega, Z, X, Y) := \omega\big(\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z\big)$ for a covector field ω.

It is the commutator of two covariant derivatives, corrected by the covariant derivative along the commutator of the fields. It is $C^\infty(M)$-linear in all of its slots, hence a tensor field (checked in the same way as for the torsion; the problem sheets do the computation of its components in terms of the connection symbols and their derivatives).

Remark (Components, not derived in the lecture). In a chart and in the convention of Lecture 7, with $R^i{}_{jmn} := \mathrm{d}x^i\big(R(\partial_j;\partial_m,\partial_n)\big)$,

$$R^i{}_{jmn} = \partial_m \Gamma^i{}_{jn} - \partial_n \Gamma^i{}_{jm} + \Gamma^i{}_{sm}\Gamma^s{}_{jn} - \Gamma^i{}_{sn}\Gamma^s{}_{jm}.$$

Geometric meaning. If one parallel transports a vector around a closed loop on a curved surface it comes back pointing in a different direction; in flat space it returns unchanged. For a small parallelogram spanned by parameter distances $\mathrm{d}\lambda$ and $\mathrm{d}\sigma$ along two directions, the net change of the vector after going around the boundary is controlled, to second order, by the Riemann tensor times the area element of the loop. This is the precise geometric meaning of curvature: an intrinsic property of the connection.

Contractions. Contracting indices of the Riemann tensor gives the *Ricci tensor* and the *scalar curvature*, coarser measures of curvature. (The lecture quotes that in a four-dimensional spacetime the Riemann tensor of the metric connection has twenty independent components; this count uses the extra symmetries of the Levi-Civita connection of a metric, not those of a general connection.)

Link to Lecture 7. The converse of the normal-coordinates theorem of Lecture 7 is stated in terms of these objects: a chart in which all $\Gamma^i{}_{jk}$ vanish throughout a neighborhood exists if and only if both torsion and curvature vanish there. Later in the course curvature is related to matter and energy via the Einstein equations, which determine the connection of spacetime.

Summary

- A vector field X is **parallel transported** along γ if $\nabla_{\dot\gamma} X = 0$, i.e. $\dot X^i + \Gamma^i{}_{jk}\dot\gamma^k X^j = 0$. The connection defines what "constant" means, and in general the result depends on the path.
- γ is **auto-parallel** if $\nabla_{\dot\gamma}\dot\gamma = 0$, i.e. $\ddot\gamma^i + \Gamma^i{}_{jk}\dot\gamma^j\dot\gamma^k = 0$. This is the connection's notion of straight line (no metric needed); the parameter is fixed up to affine transformations. On the round sphere the solutions are great circles.
- The **torsion** $T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]$ is a tensor whose components are the antisymmetric part of Γ; we use torsion-free connections.
- The **Riemann curvature** $R(Z;X,Y) = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z$ is a tensor that measures the rotation of a vector transported around an infinitesimal loop; contractions give the Ricci tensor and scalar curvature.